

Status update (8 October 2026). This summary covers notebooks 01–09 (runs 5/6). Later work changed some conclusions: **withdrawn** — perivascular T cells orienting 18S towards vessels (notebook 19: bleed from bright neighbours + outline geometry), a vimentin "ring" at lesion edges and vimentin-border containment (notebooks 16–17); **downgraded** — 18S texture as a severity marker (replicates only in glia/myeloid, notebook 13); **replicated** — white-matter neuropil loss, astrocyte RNA before vimentin protein, image-only models on unseen runs (notebook 13). Current overall verdict: [report/BeyondBoundaries_verdict.pdf](#) ; disease biology: [report/BeyondBoundaries_lesion_story.pdf](#) .

Beyond Boundaries — what do the Xenium multimodal segmentation stains add to the transcriptome?

Data. Mouse EAE spinal cord (RRMAP2 runs 5 and 6): 18 annotated Xenium 5K sections, 25 animals (chronic and relapse-remitting EAE, CFA controls), 500,379 annotated cells. Four morphology channels besides transcripts: DAPI · ATP1A1/CD45/E-Cadherin (boundary) · 18S rRNA (interior RNA) · αSMA/Vimentin (interior protein).

Approach. Per cell and channel: intensities in 6 compartments (nucleus, cytoplasm, 1 μm rim, 2 μm ring, 10 μm territory), radial profile, polarity, texture, morphology (201 features; notebook 01) → background / normalisation (02) → do the stains show known biology (03) → how much of each image feature does the transcriptome explain, and vice versa (04) → is the rest technical or biological; does it add per animal (05) → targeted readouts (06) → look at the cells (07).

This notebook only *collects* results (figures from [results/](#) , tables from CSVs); every number links to the notebook that computed it.

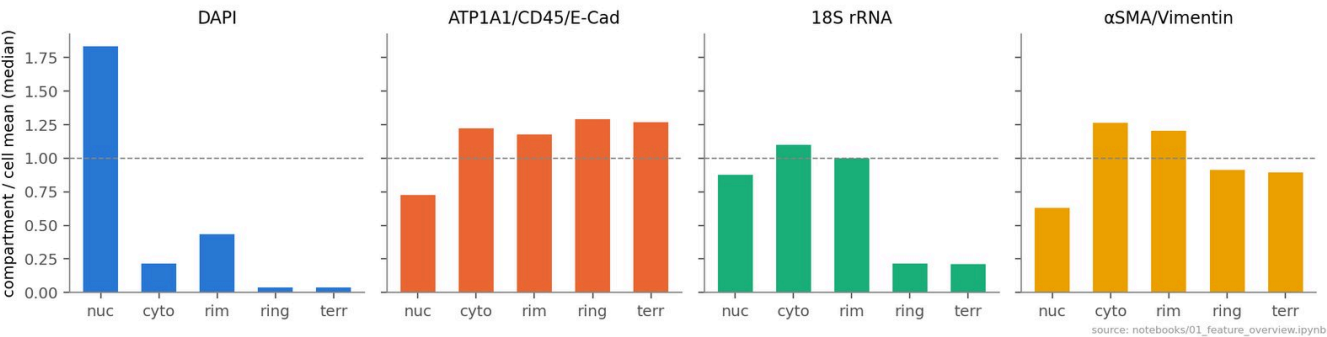
Bottom line

question	answer	where
Do the stains show known biology?	Yes — VSMC αSMA, leukocyte & ependymal vimentin, neuron 18S, low ch3 in neurons/oligos; all 25 animals	03
Is CD45 usable?	No — boundary channel behaves as ATP1A1 (neuropil) only, even in low-neuropil areas	03
How much of the image does the transcriptome explain?	~15 % per feature (median); morphology and rim/cyto intensities best	04

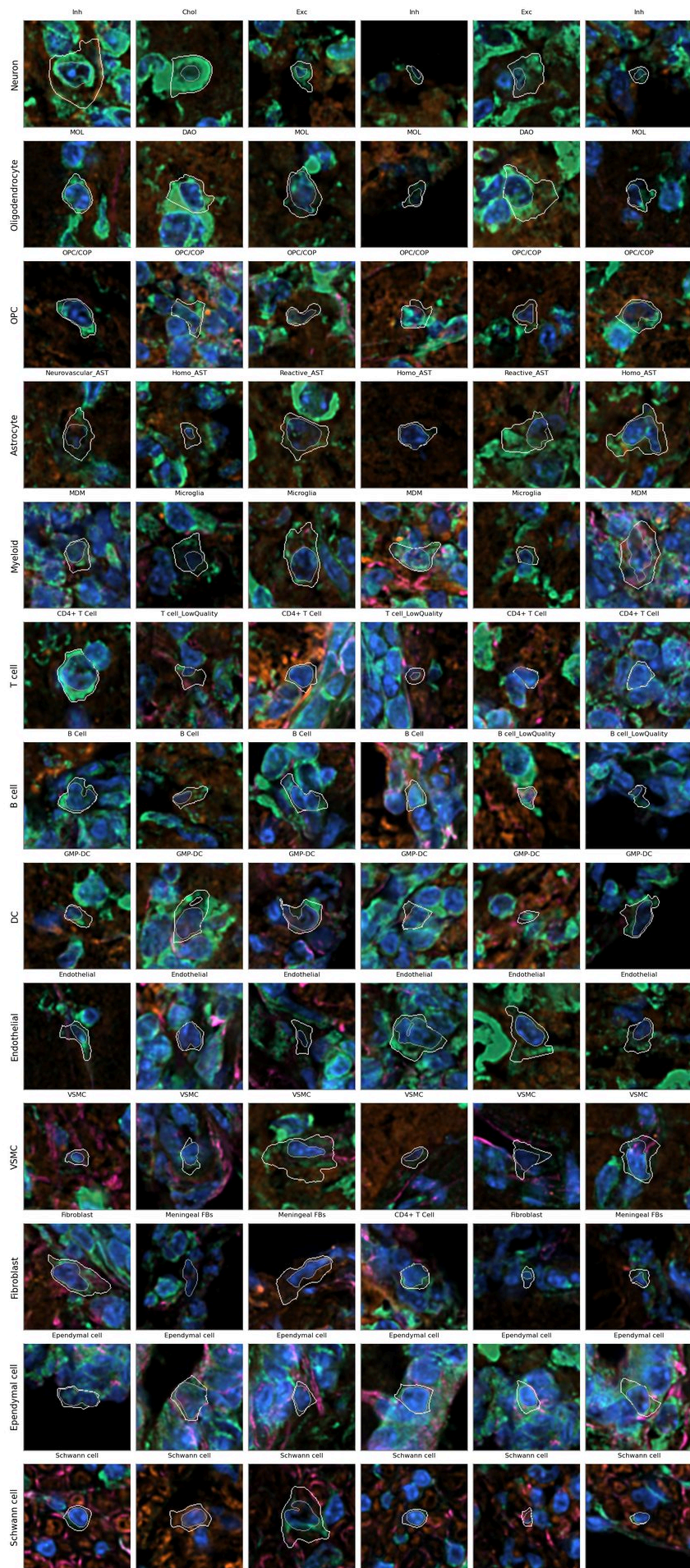
question	answer	where
Is the unexplained ~85 % hidden biology?	Mostly not — a local optical/staining field shared across cell types + cell-level variation unrelated to lesions	05
Do images add to the transcriptome for lesion identity?	No ($\leq +0.015$ AUROC; niches are transcriptome-defined)	04
...for clinical score per animal?	Images alone $p = 0.46$ ($p = 0.04$) vs transcriptome 0.86; combined not better	05
What can images predict <i>alone</i> ?	Cell type 46 % (14 types, chance 7 %), subtypes (microglia 0.72, reactive astro 0.68), anatomy 63 % (10 regions), lesion maps AUROC 0.90 , clinical score $p = 0.62$ — always $\leq$ transcriptome, but images add to it for anatomy ($0.68 \rightarrow 0.73$)	08
What <i>do</i> they add?	Measurements of genes not on the panel (<i>Vim</i> , <i>Acta2</i> , <i>Atp1a1</i>): vimentin reactivity timing (replicated), WM neuropil loss (–21 to –28 %, replicated) , 18S gain in activated lesion cells; an 18S-texture severity marker only partly replicated (glia/myeloid). T-cell 18S polarity towards vessels withdrawn (artefact)	05–07, 13, 19

1. Features look right (notebook 01)

Mask areas equal Xenium's exactly; centroids within 0.28 μm ; 0 truncated cells. Each stain has its own compartment — notably **ATP1A1 is higher *outside* cells** (neuropil).

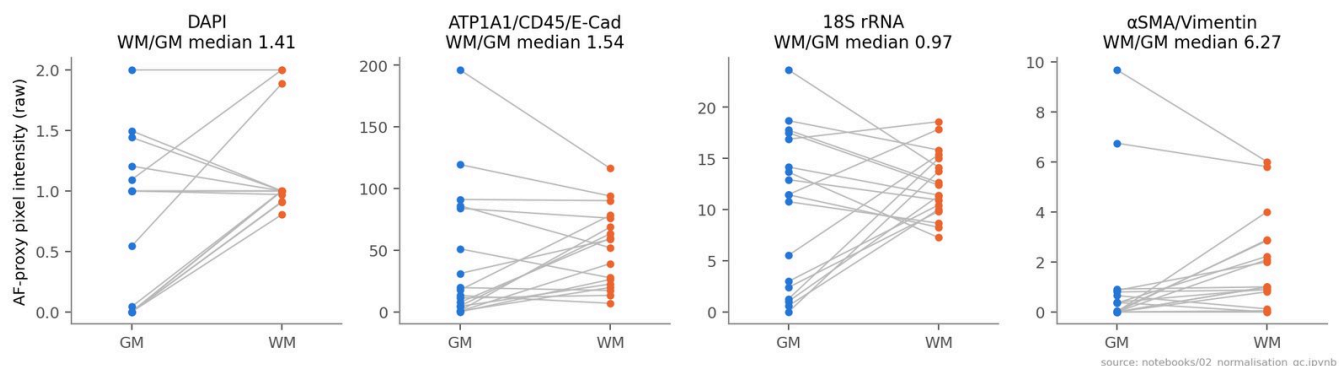


results/01_features/compartment_profile.png

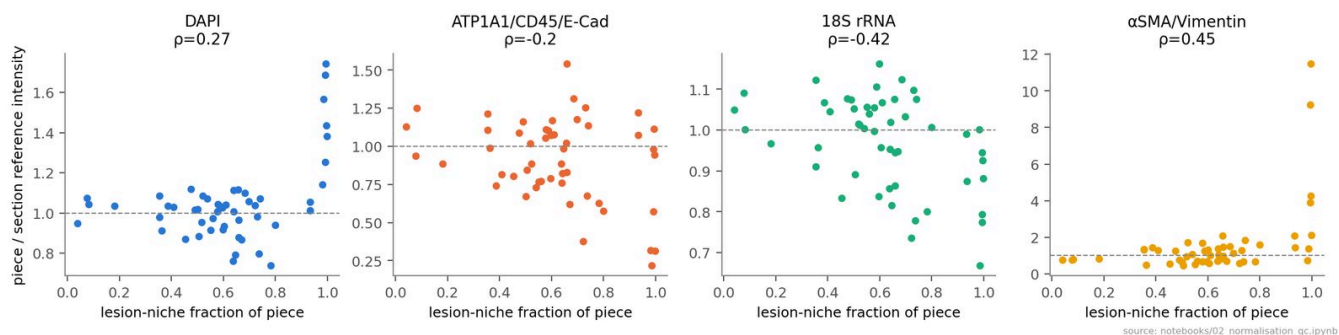


2. Background and normalisation (notebook 02)

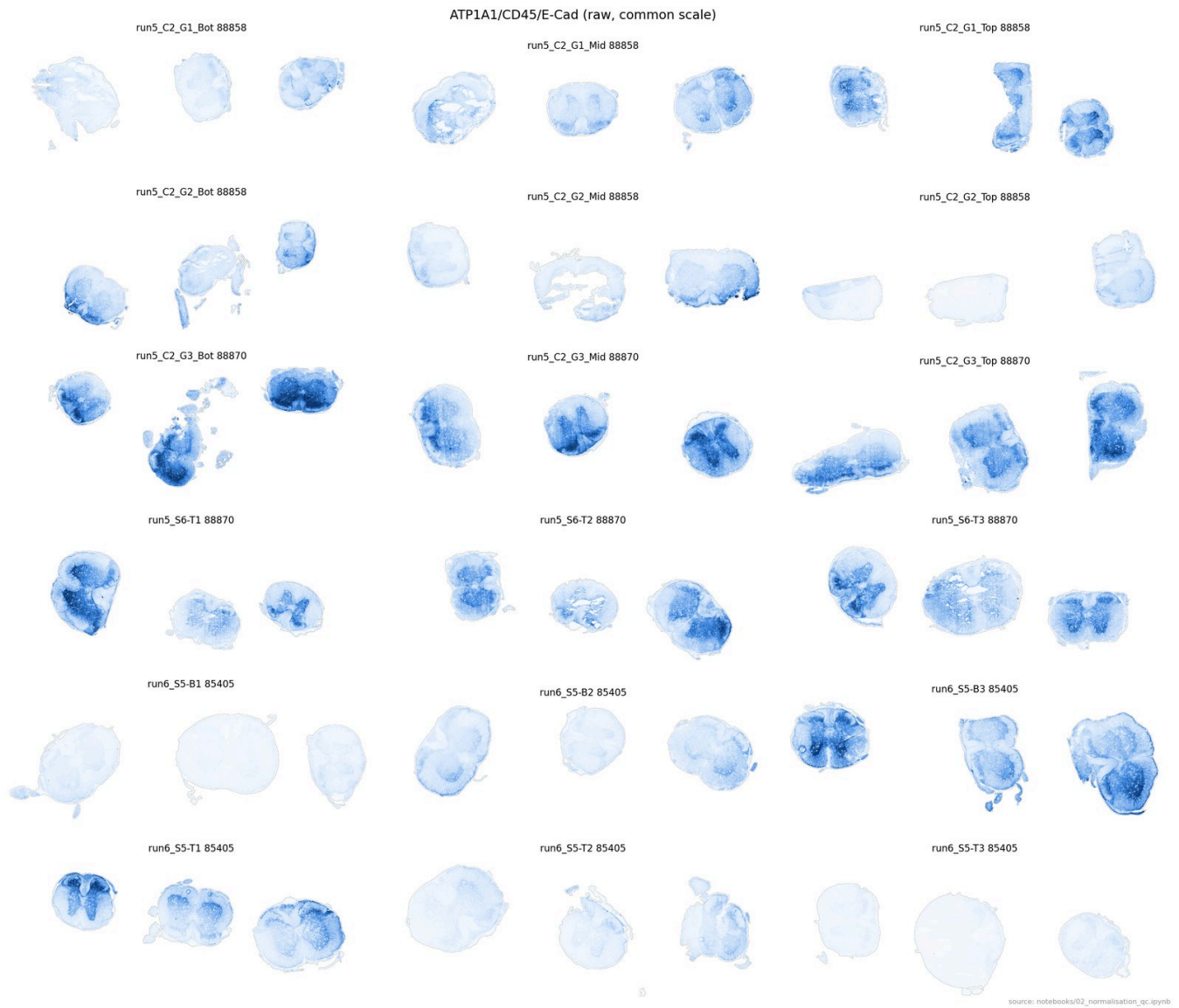
No broadband myelin autofluorescence (18S WM/GM 0.97). Final normalisation = local background subtraction, scaled per section by the p90 of physiological-niche cells. **Stain intensity tracks disease at piece level:** α SMA/Vim brighter ($\rho = 0.45$, $p = 8e-4$) and 18S dimmer ($\rho = -0.42$, $p = 0.002$) in physiological-niche cells of sicker animals.



results/02_normalisation_qc/af_wm_vs_gm.png



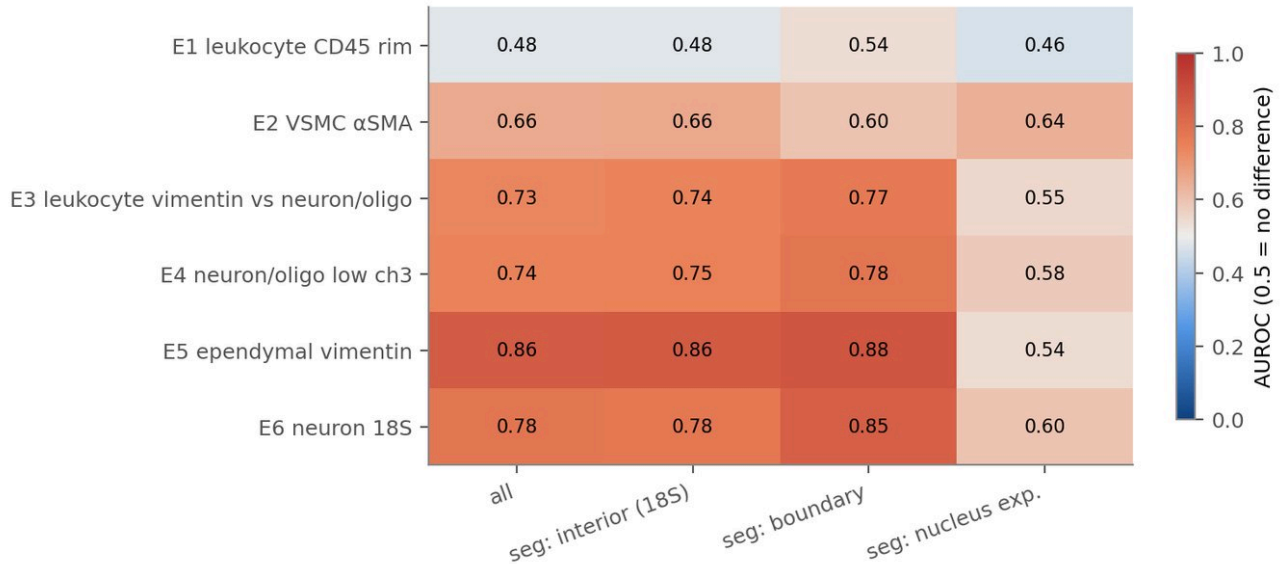
results/02_normalisation_qc/piece_intensity_vs_lesion.png



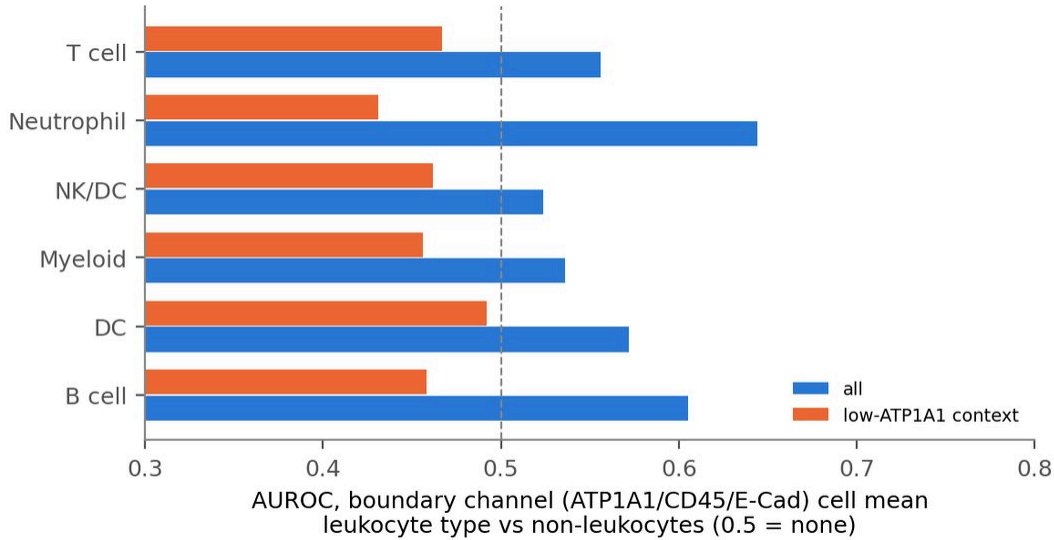
results/02_normalisation_qc/spatial_bnd.png

3. The stains recover known biology — except CD45 (notebook 03)

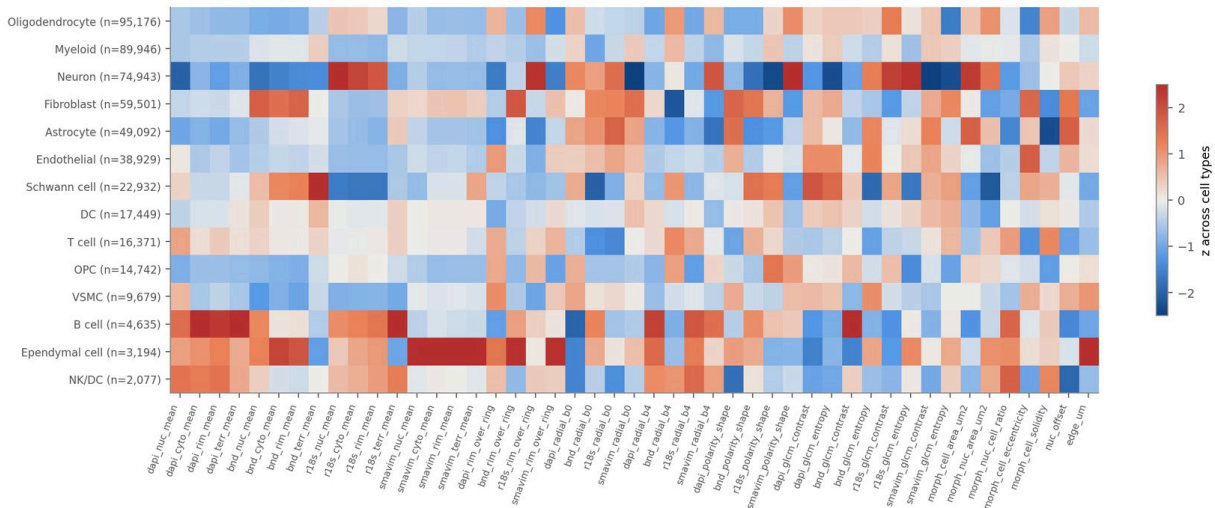
	test	feature	auroc	verdict
0	E1 leukocyte CD45 rim	bnd_rim_over_ring	0.477	fails
5	E2 VSMC α SMA	smavim_cell_mean	0.656	holds
10	E3 leukocyte vimentin vs neuron/oligo	smavim_cell_mean	0.734	holds
15	E4 neuron/oligo low ch3	smavim_cell_mean	0.743	holds
20	E5 ependymal vimentin	smavim_cell_mean	0.859	holds
25	E6 neuron 18S	r18s_cyto_mean	0.780	holds



results/03_sanatlas/expectation_tests.png

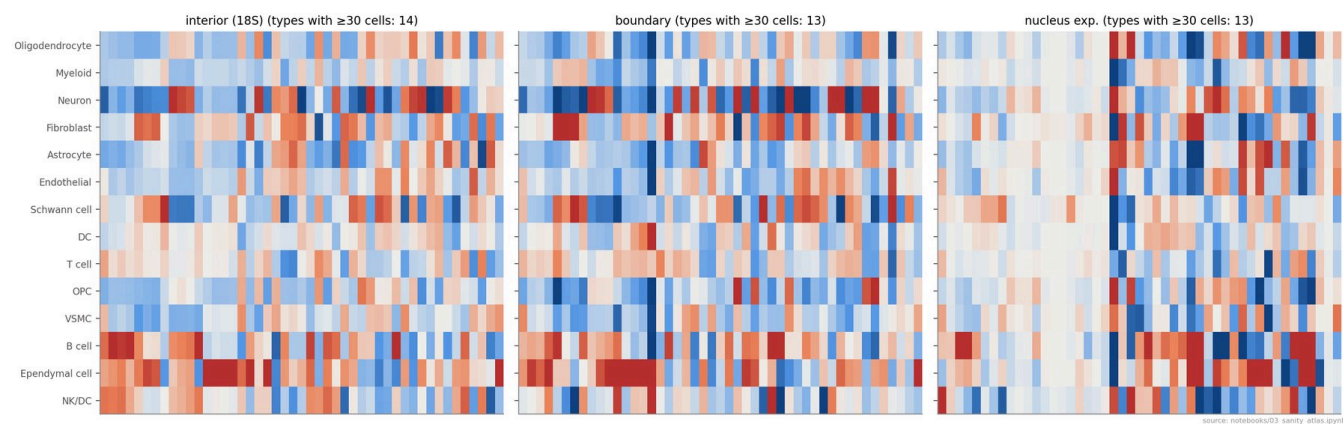


results/03_sanatlas/E1_cd45_context.png



results/03_sanity_atlas/atlas_celltype_heatmap.png

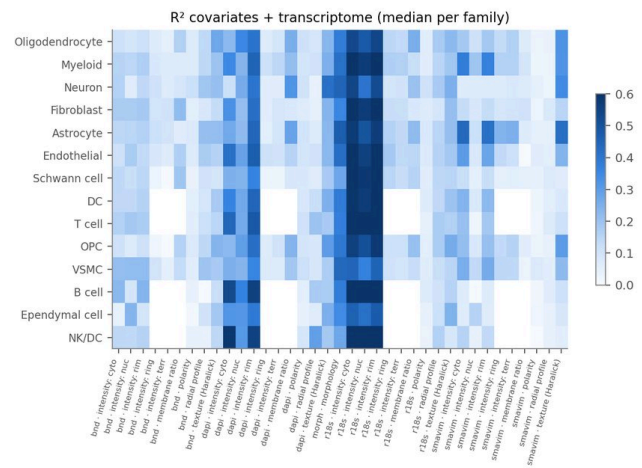
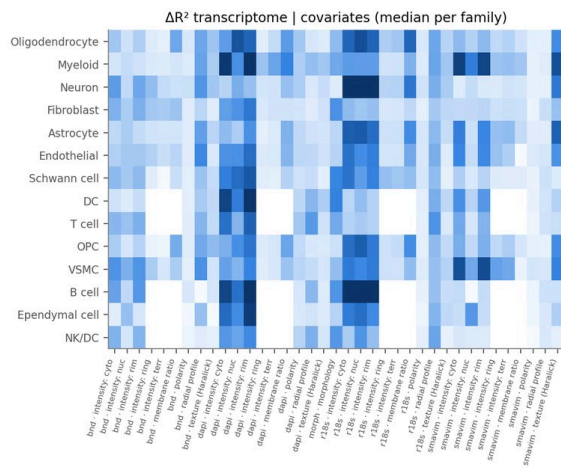
Segmentation method shifts shape/distribution features for every cell type (the stain drew that edge), so all downstream analyses use 18S-segmented cells (96 %).



results/03_sanity_atlas/atlas_by_segmethod.png

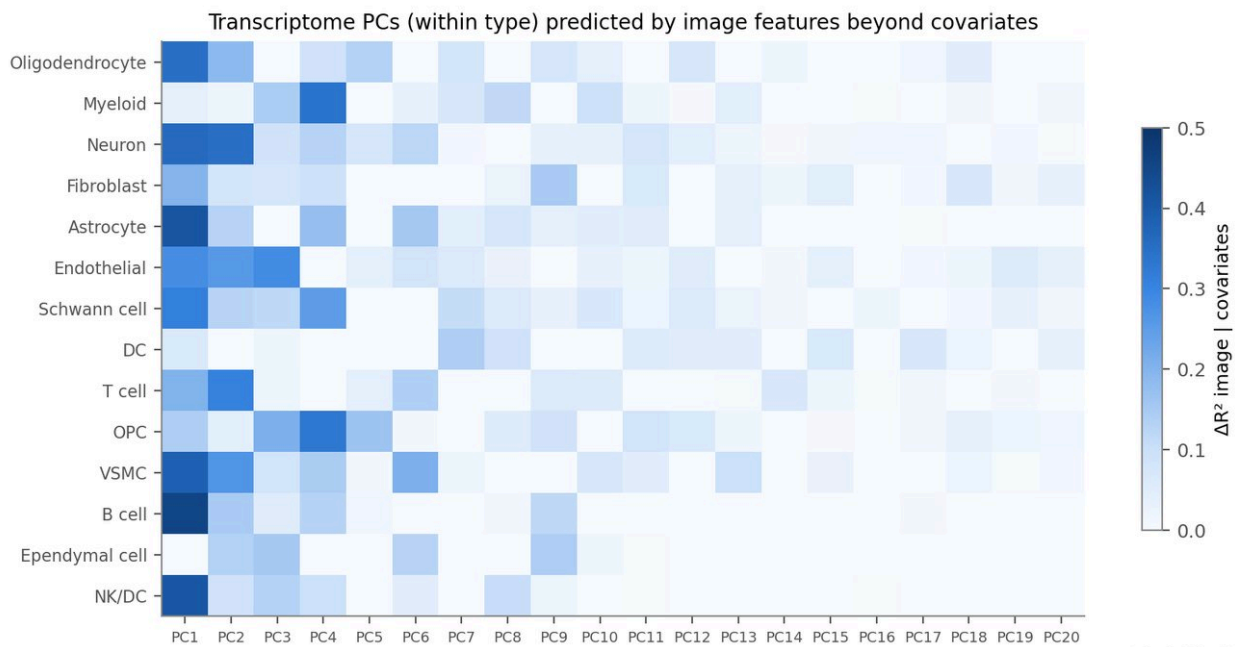
4. The transcriptome explains ~15 % of the image; images predict transcriptional state (notebook 04)

	R ² covariates	R ² covariates + transcriptome	ΔR ² transcriptome
cell_type			
Schwann cell	0.029	0.112	0.068
Fibroblast	0.042	0.123	0.068
Neuron	0.035	0.127	0.072
Ependymal cell	0.019	0.130	0.064
NK/DC	0.068	0.142	0.056
VSMC	0.037	0.148	0.091
B cell	0.053	0.148	0.067
Oligodendrocyte	0.048	0.152	0.090
Endothelial	0.063	0.156	0.075
OPC	0.048	0.160	0.067
Myeloid	0.055	0.163	0.105
DC	0.048	0.165	0.080
T cell	0.052	0.171	0.091
Astrocyte	0.081	0.177	0.071



source: notebooks/04_orthogonality.ipynb

results/04_orthogonality/r2_heatmaps.png



source: notebooks/04_orthogonality.ipynb

results/04_orthogonality/reverse_pcs.png

Genes best predicted by the images (beyond covariates), top 6 per cell type — reactive astrocyte genes, neurofilaments, oligodendrocyte maturation genes, and neighbour spill-over genes:

top genes predicted by image

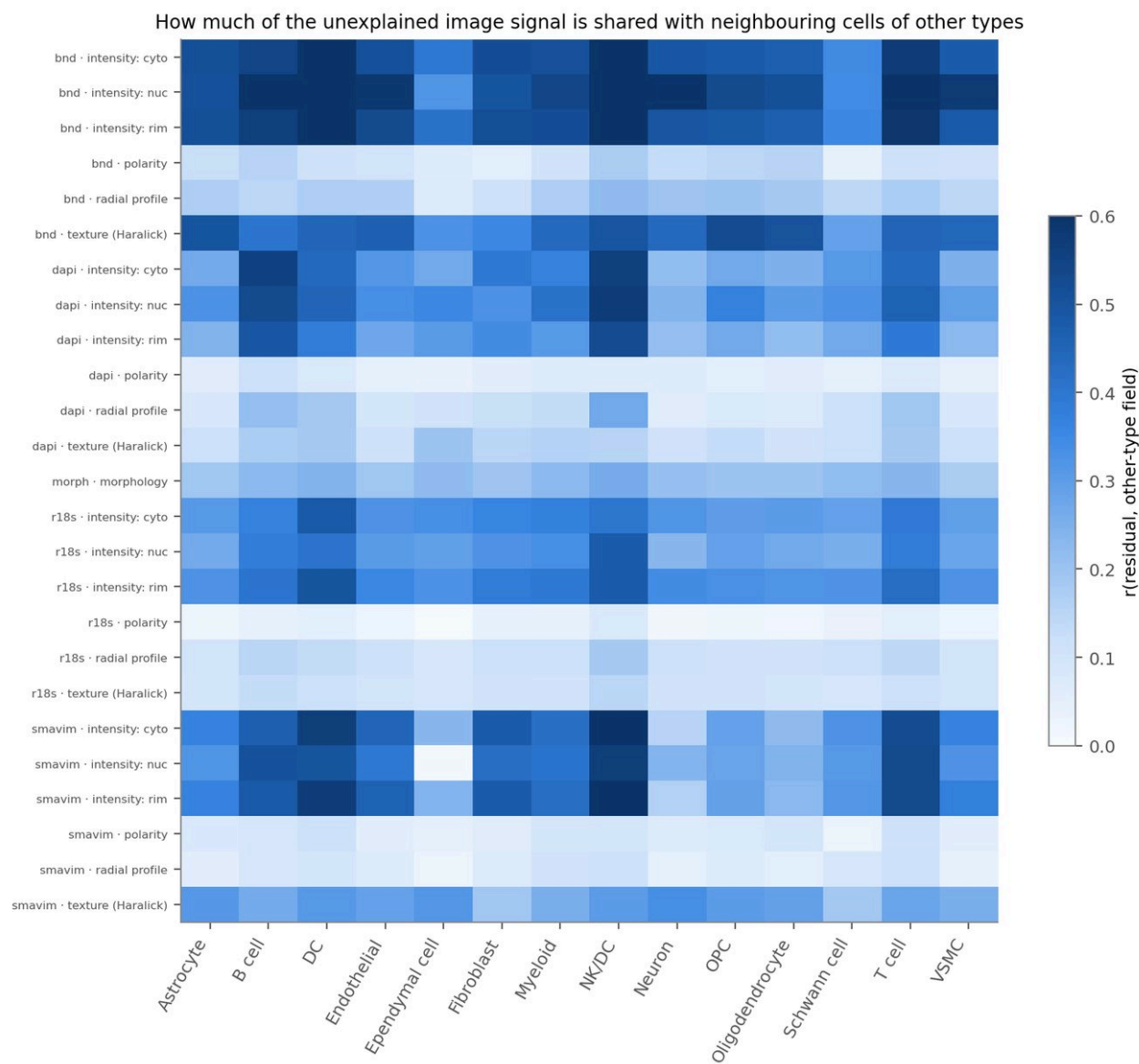
cell_type	
Astrocyte	Slc6a11, C3, Snap25, Fgfr3, A2m, Aldoc
B cell	Jchain, Igkc, Sel1l, Gfap, Tent5c, Mzb1
DC	Mbp, Neat1, Gfap, Snap25, Cst7, Lamp1
Endothelial	Snap25, Eno2, Aldoc, Ckb, Sncb, Map2
Ependymal cell	Mbp, Adamts20, Syne1, Plec, Tll6, Rsph1
Fibroblast	Gfap, Ptgs, Mbp, Neat1, Ctss, Crabp2
Myeloid	Aldoc, Snap25, Map2, Ckb, Nefl, Mbp
NK/DC	Mbp, Gfap, Il2rb, Klrk1, Cd74, H2-D1
Neuron	Meg3, Mbp, Nefm, Nefh, Map1b, Snap25
OPC	Gfap, Snap25, Aldoc, Sncb, Eno2, Glra1
Oligodendrocyte	Snap25, Gfap, Klk6, Eno2, Sncb, Ermn
Schwann cell	Pmp22, Cnp, Mbp, Cryab, Smtn, Cd9
T cell	Mbp, Gfap, H2-D1, C3, Aldoc, Cd74
VSMC	Vtn, Pdgfrb, Snap25, Mbp, Aldoc, Kcnj8

Images are redundant with the transcriptome for lesion-niche identity:

	n	lesion_frac	AUROC tx	AUROC tx+img	AUROC img	Δ AUROC (img tx)
cell_type						
Oligodendrocyte	29895	0.605	0.928	0.929	0.650	0.001
Myeloid	29860	0.942	0.932	0.929	0.670	-0.003
Neuron	29988	0.290	0.846	0.850	0.471	0.004
Fibroblast	29183	0.916	0.884	0.898	0.716	0.015
Astrocyte	29816	0.603	0.947	0.947	0.757	-0.000
Endothelial	27406	0.645	0.937	0.938	0.747	0.001
Schwann cell	20610	0.367	0.936	0.940	0.769	0.004
OPC	14183	0.680	0.955	0.958	0.822	0.003
VSMC	8375	0.791	0.794	0.807	0.650	0.013
Ependymal cell	3005	0.062	0.948	0.946	0.816	-0.002

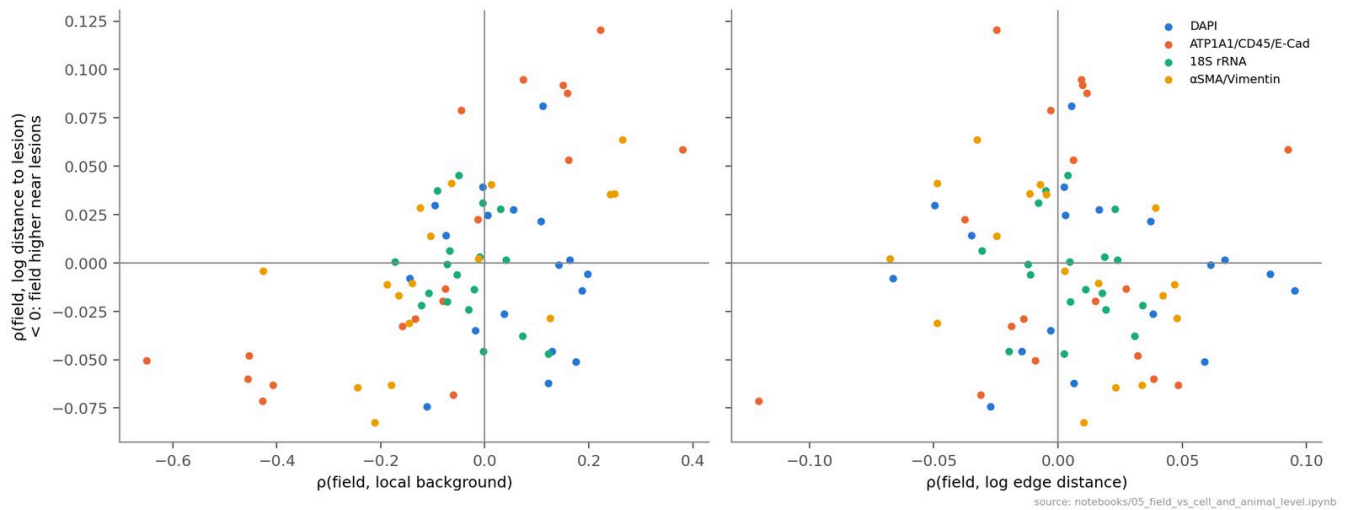
5. What is the unexplained 85 %? (notebook 05)

Much of it is shared with neighbouring cells of *other* types (a local field) and that field does not follow lesions; the cell-specific remainder barely does either.



source: notebooks/05_field_vs_cell_and_animal_level.ipynb

results/05_field_and_animal/field_share_heatmap.png



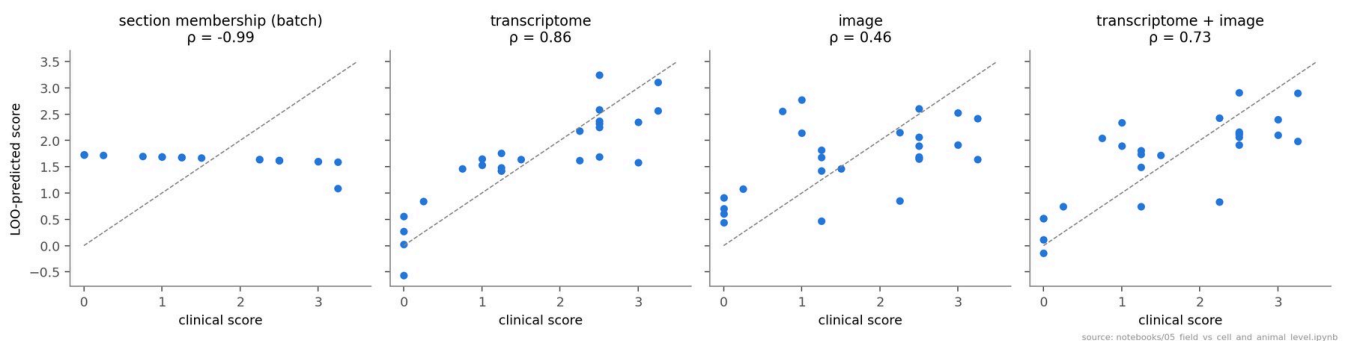
results/05_field_and_animal/field_lesion_vs_technical.png

tests $q < 0.05$

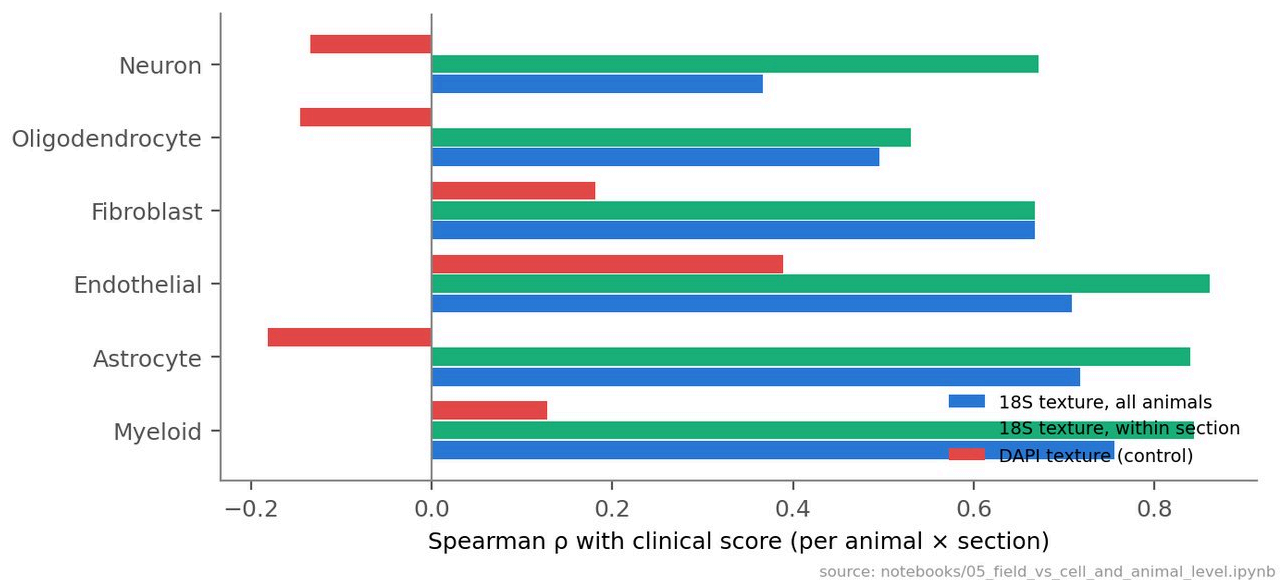
part		
cell-specific	924	19
field	924	21
full residual	924	19

6. Per animal: images carry real but weaker disease information; 18S texture (later only partly replicated, notebook 13)

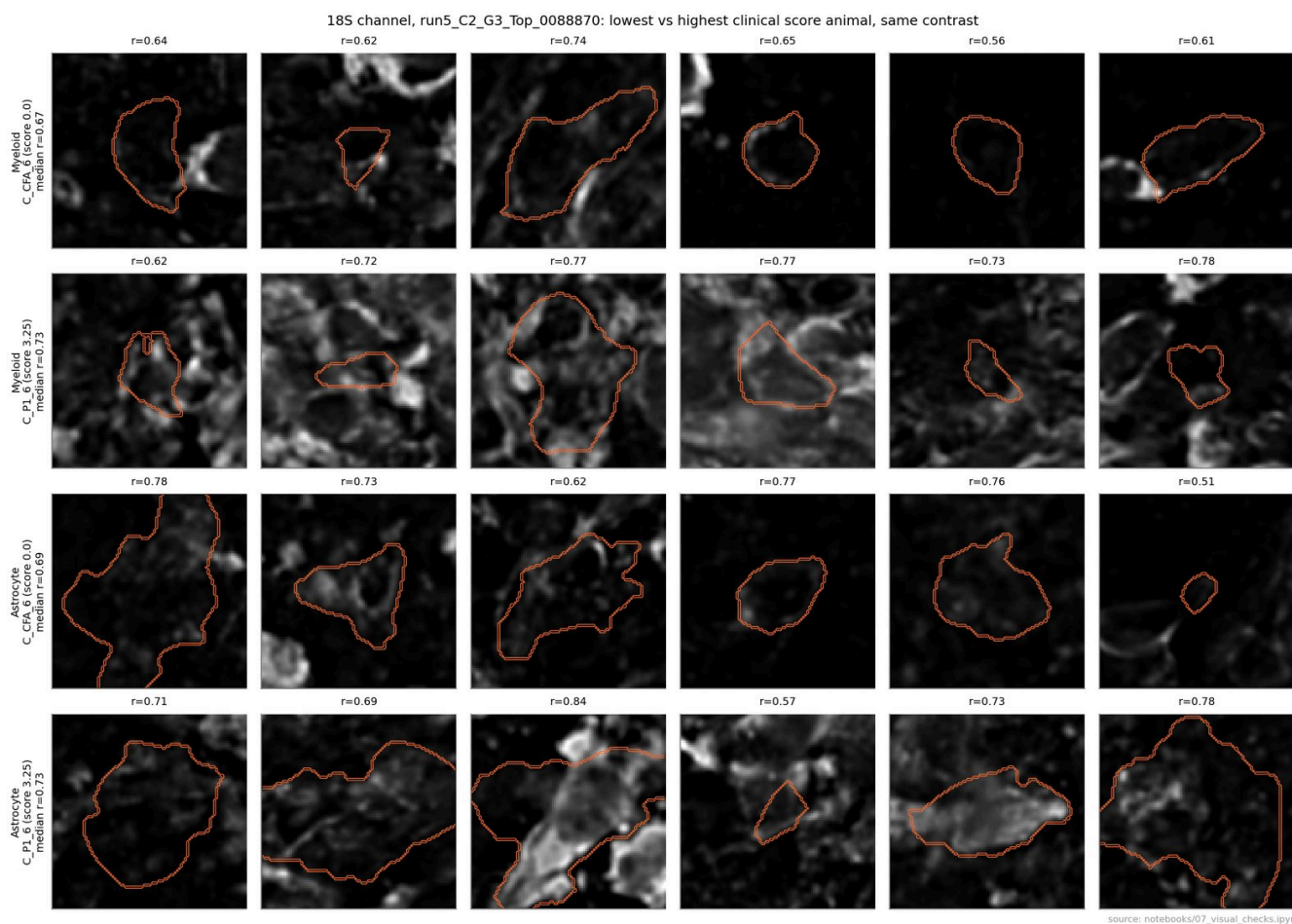
	R^2	Spearman ρ	perm p (ρ)
section membership (batch)	-0.152	-0.990	NaN
transcriptome	0.748	0.859	0.005
image	0.284	0.456	0.040
transcriptome + image	0.586	0.735	NaN



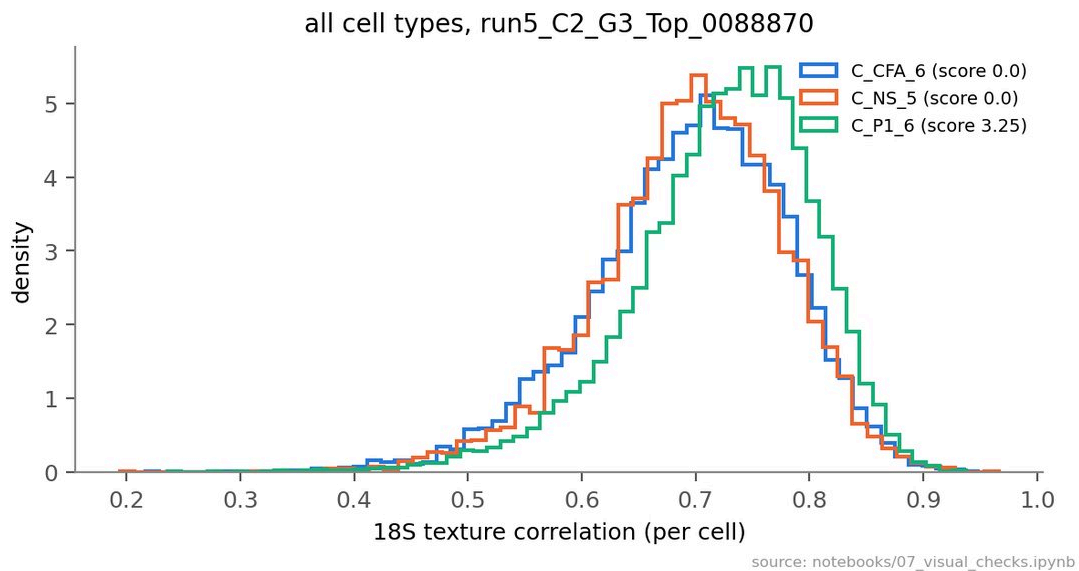
results/05_field_and_animal/animal_score_prediction.png



results/05_field_and_animal/r18s_texture_within_section.png

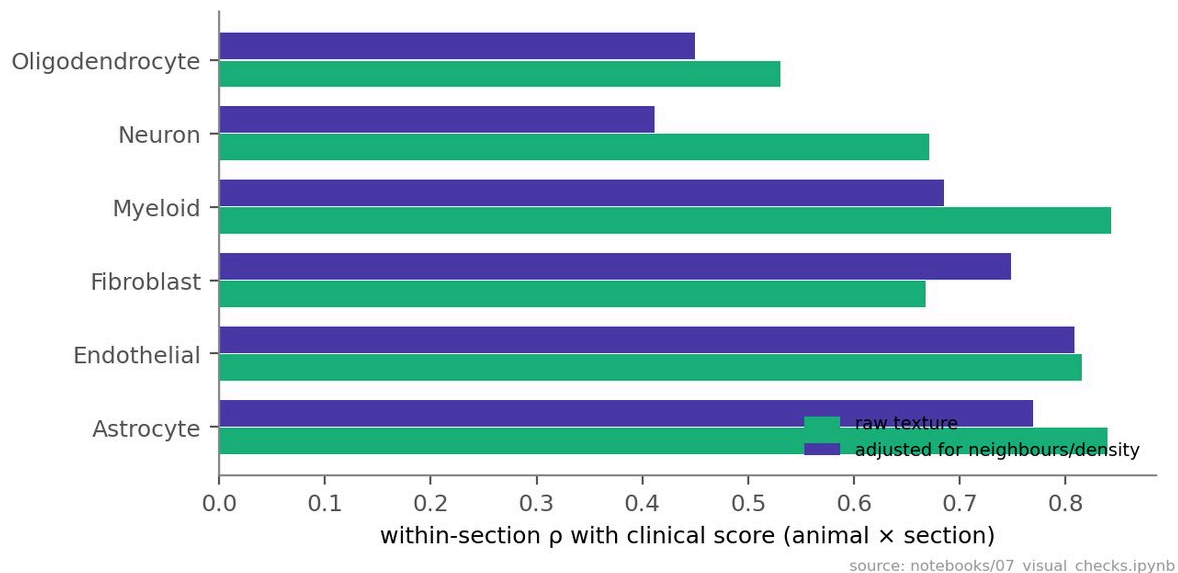


results/07_visual_checks/r18s_texture_low_vs_high_score.png



results/07_visual_checks/r18s_texture_per_animal_hist.png

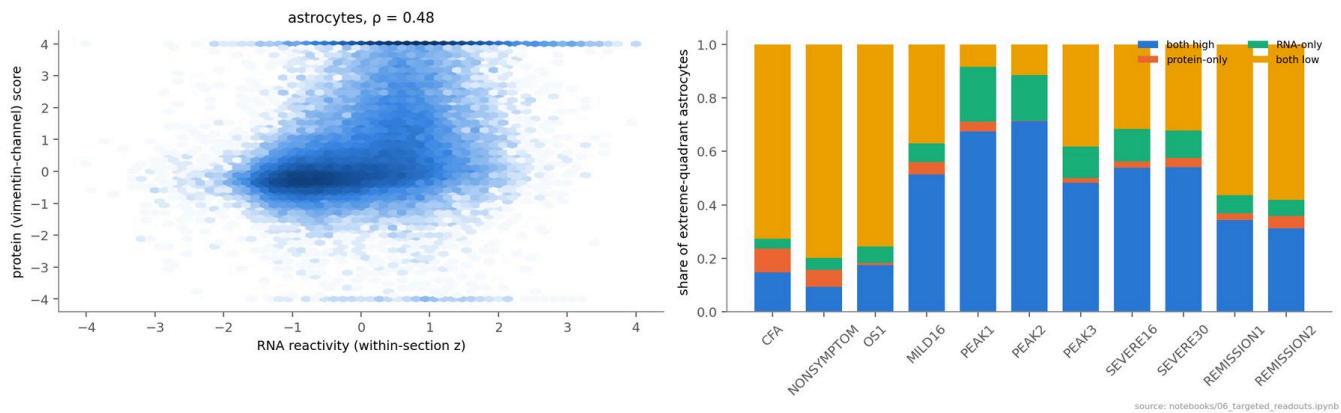
Is it just crowding (bright 18S neighbours bleeding in)? Mostly not — the within-section link survives adjustment for neighbouring 18S, local density and cell size:



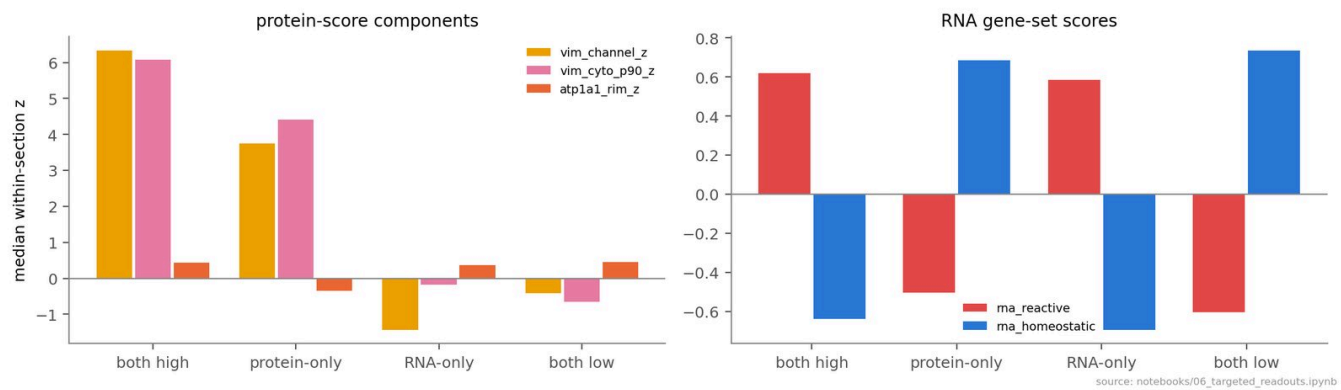
results/07_visual_checks/r18s_texture_crowding_adjustment.png

7. Targeted readouts (notebooks 06, 07)

Astrocyte reactivity: vimentin protein vs RNA — transcription comes first

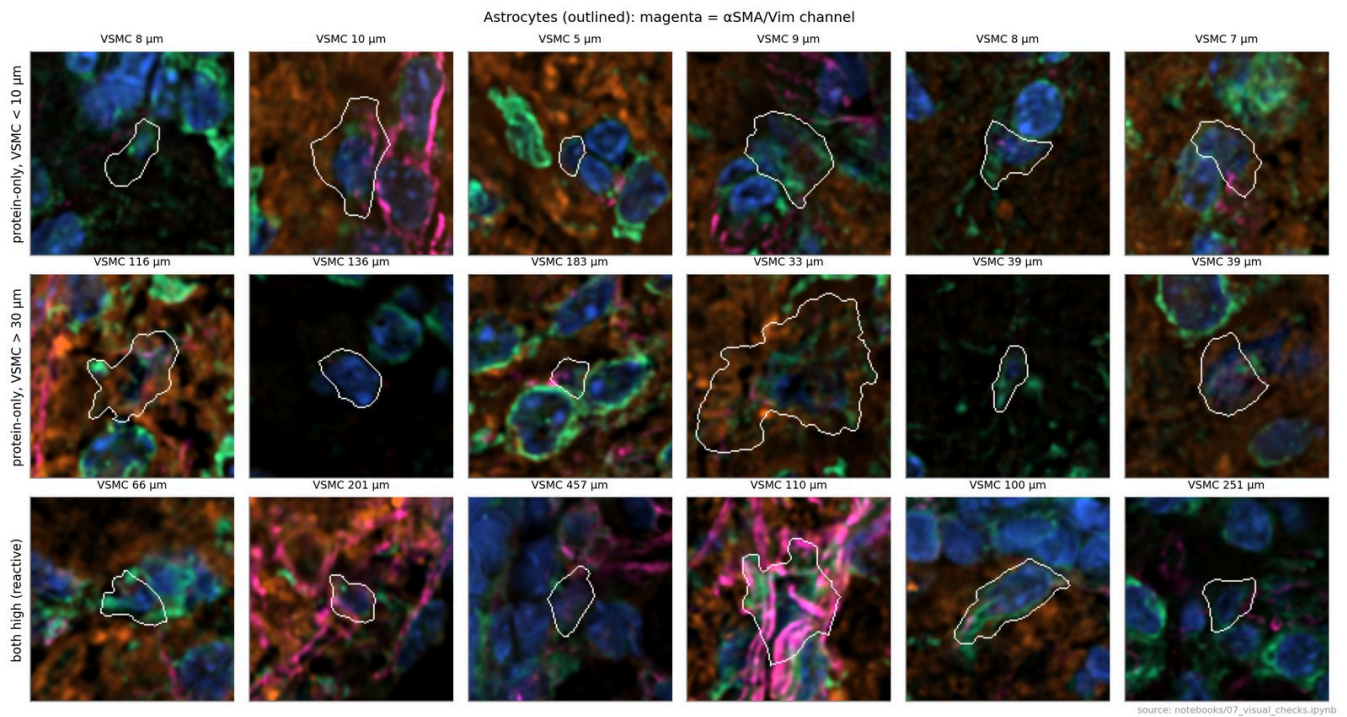


results/06_targeted_readouts/astro_protein_vs_rna.png



results/06_targeted_readouts/astro_quadrant_components.png

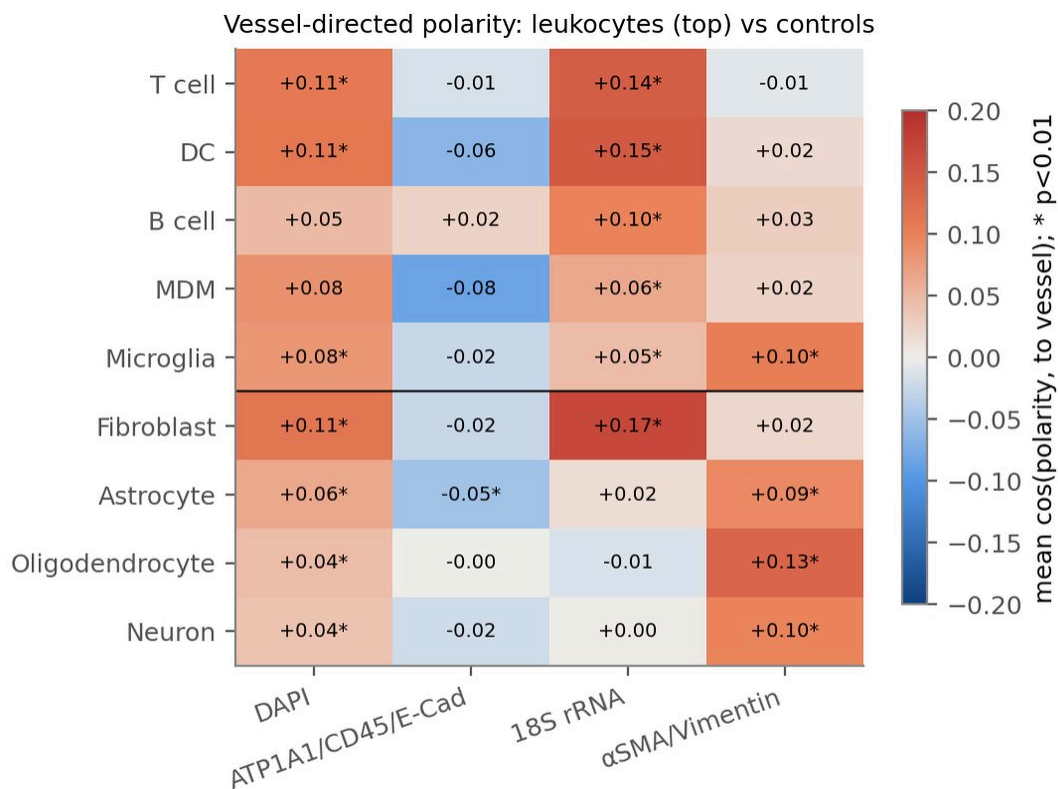
	n	lesion_frac	WM_frac	lesion_dist_median	edge_um_median	area_median	trans
quadrant							
both high	5853	0.950	0.856	0.000	101.389	88.190	
protein-only	401	0.284	0.147	29.779	297.379	74.011	
RNA-only	1312	0.949	0.772	0.000	127.557	68.141	
both low	4833	0.137	0.115	33.896	443.296	82.907	



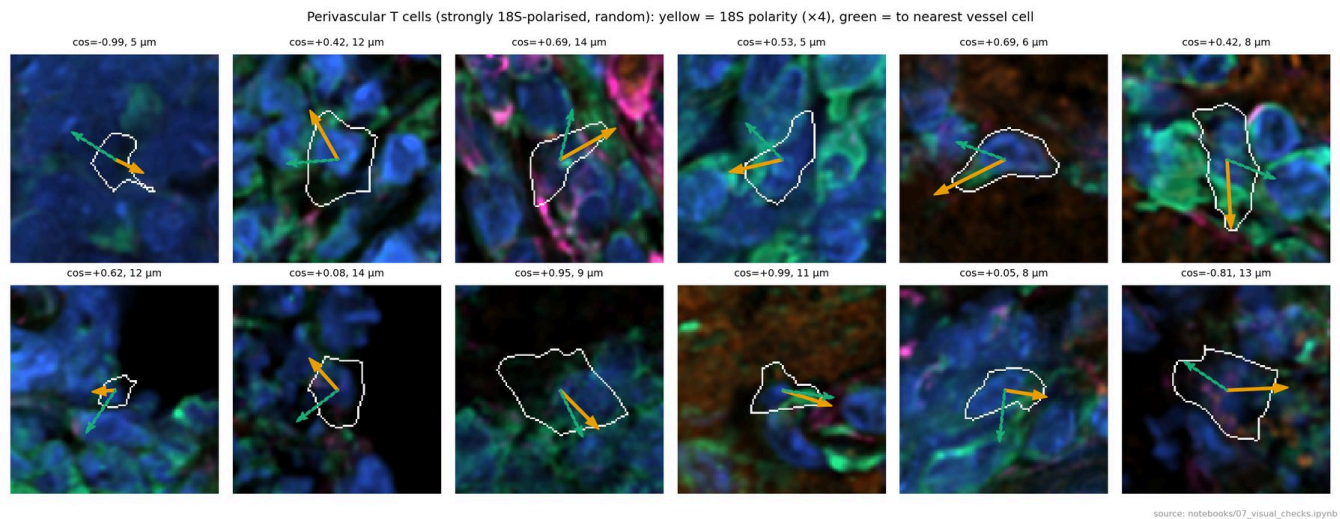
results/07_visual_checks/astro_protein_only_vsmc_gallery.png

Leukocyte polarity — WITHDRAWN (notebook 19)

The 18S "towards vessels" signal below is bleed from 18S-bright neighbours plus outline geometry (a brightness-matched control removes it). The figures are kept for the record only.

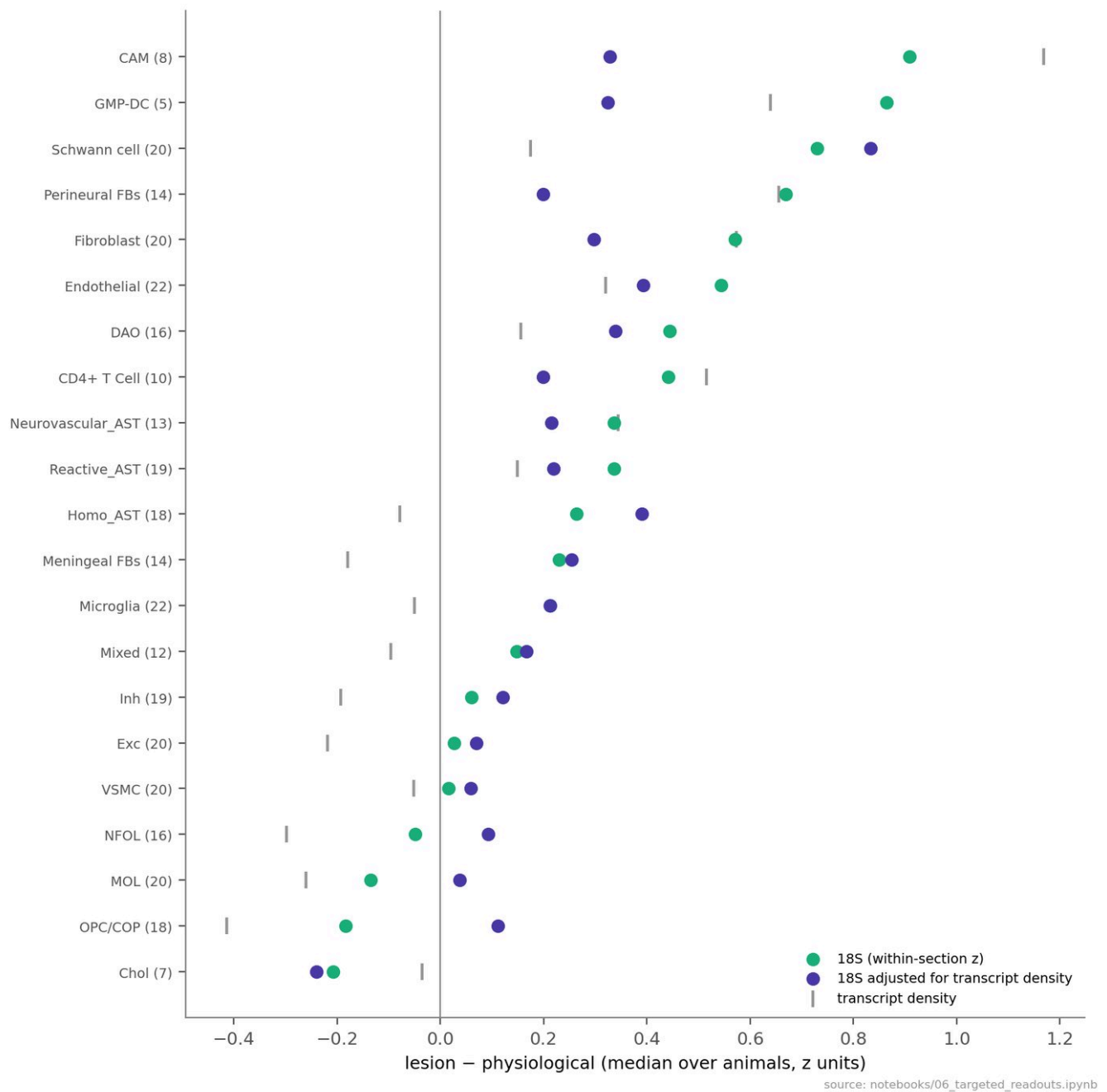


results/06_targeted_readouts/polarity_vessel_direction.png



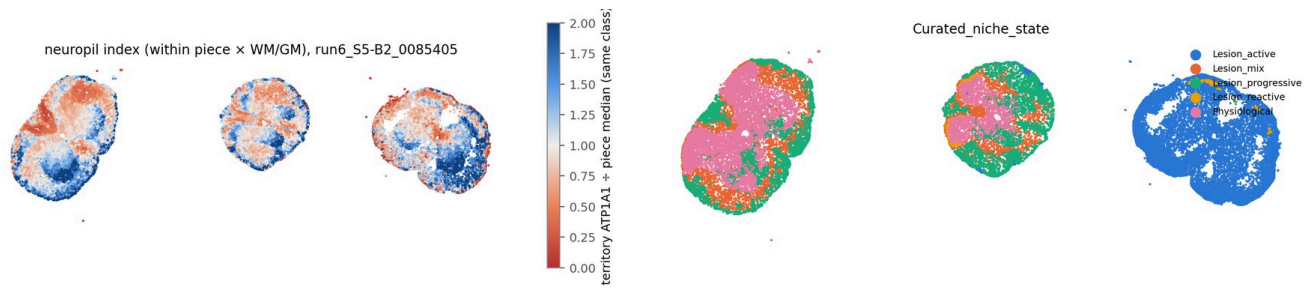
results/07_visual_checks/tcell_perivascular_polarity.png

18S rises in activated cells in lesions, beyond RNA content



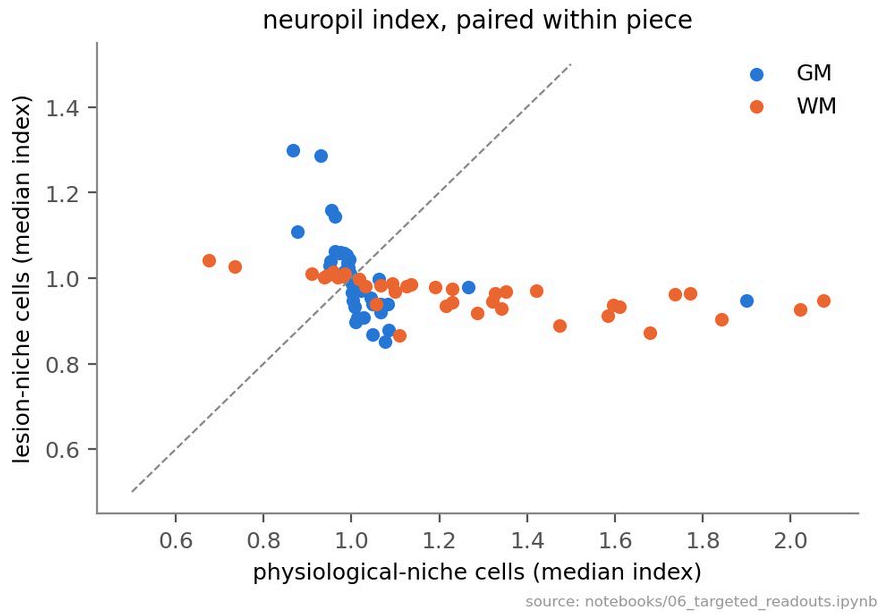
results/06_targeted_readouts/r18s_lesion_by_type.png

Neuropil loss: –25 % surrounding ATP1A1 in white-matter lesions (within piece)



source: notebooks/06_targeted_readouts.ipynb

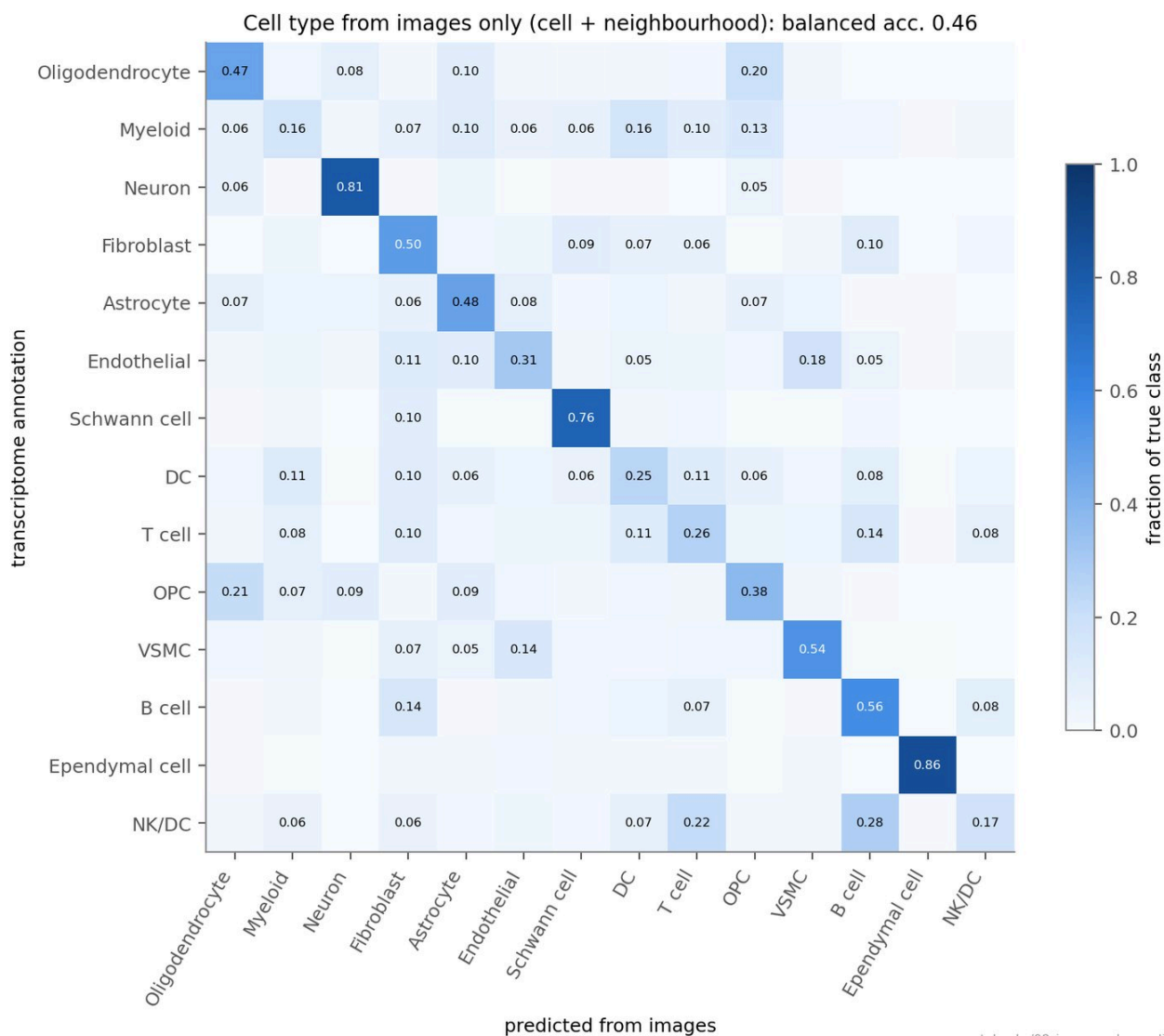
results/06_targeted_readouts/neuropil_index_map.png



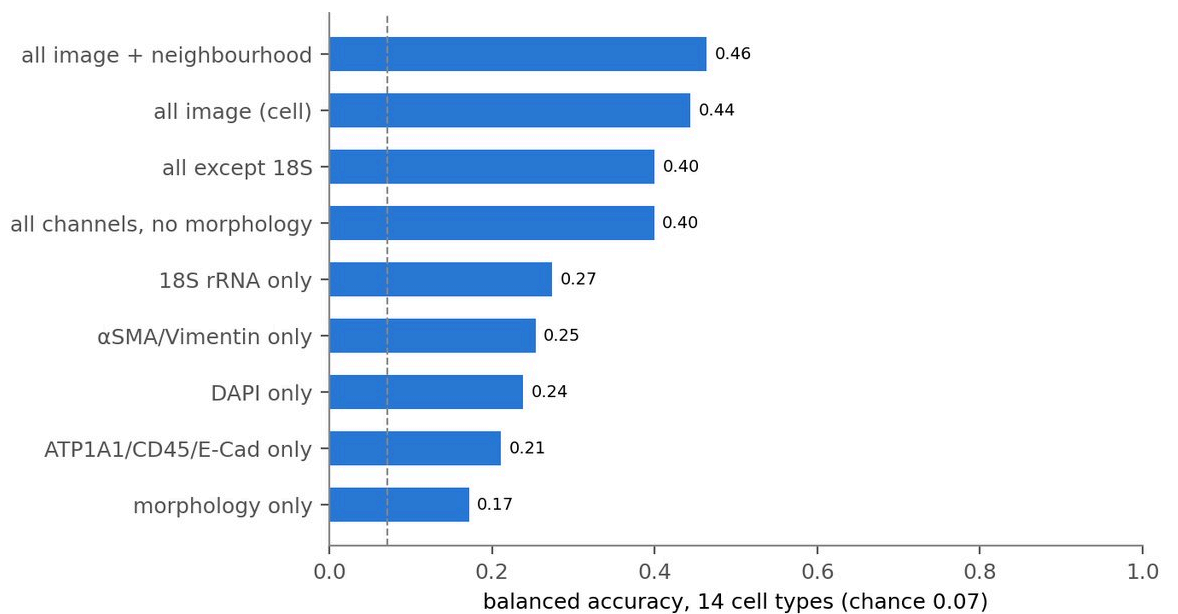
results/06_targeted_readouts/neuropil_index_paired.png

8. What the images can predict on their own (notebook 08)

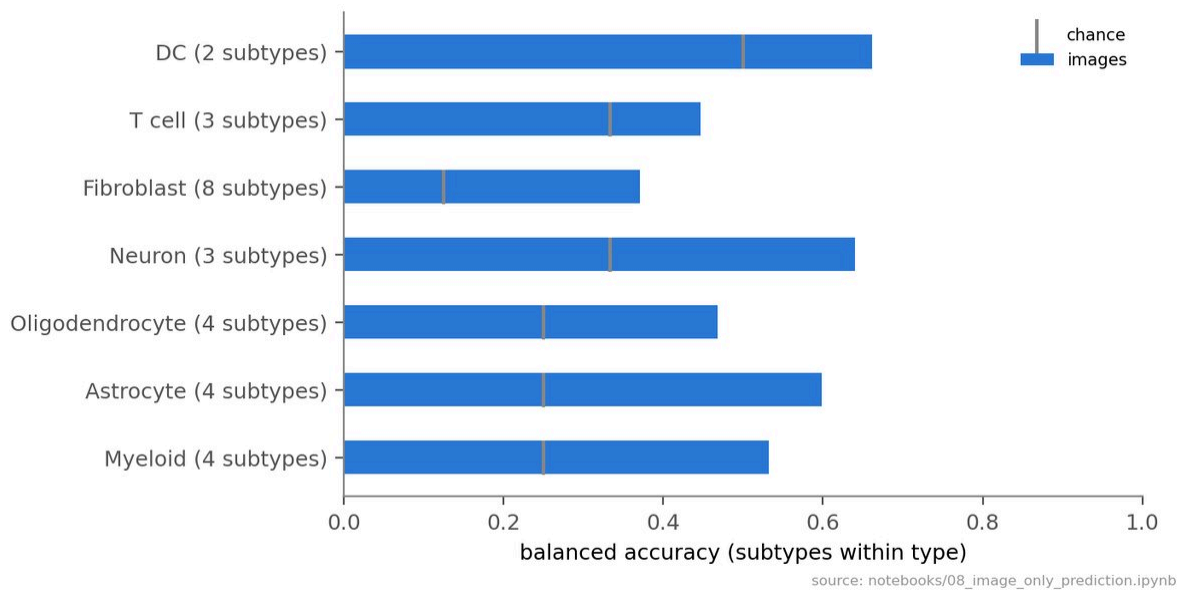
Gradient-boosted trees on image features only, cross-validated by animal. Cell type at 46 % balanced accuracy (14 types, chance 7 %), errors within lineages; no single stain carries identity; subtypes recoverable.



results/08_image_only_prediction/L1_confusion.png



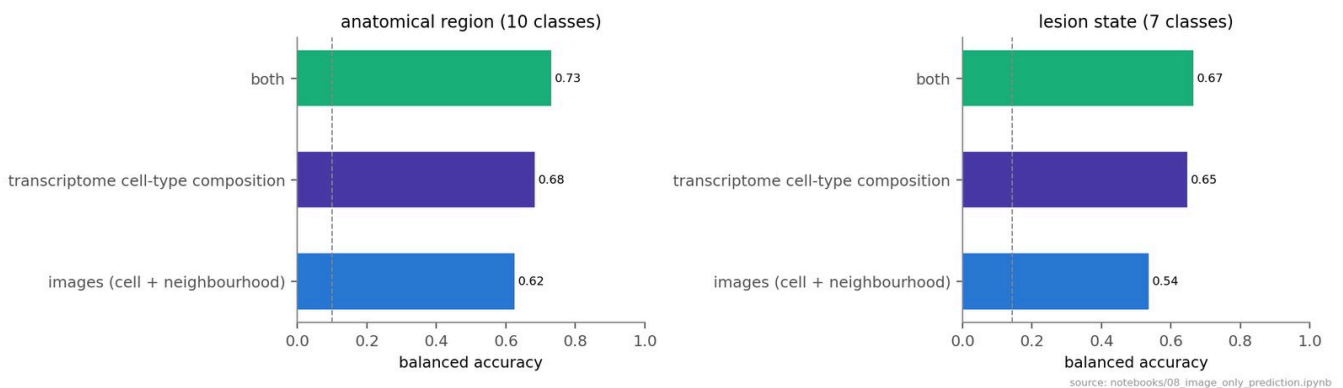
results/08_image_only_prediction/L1_feature_set_ablation.png



results/08_image_only_prediction/L2_within_type.png

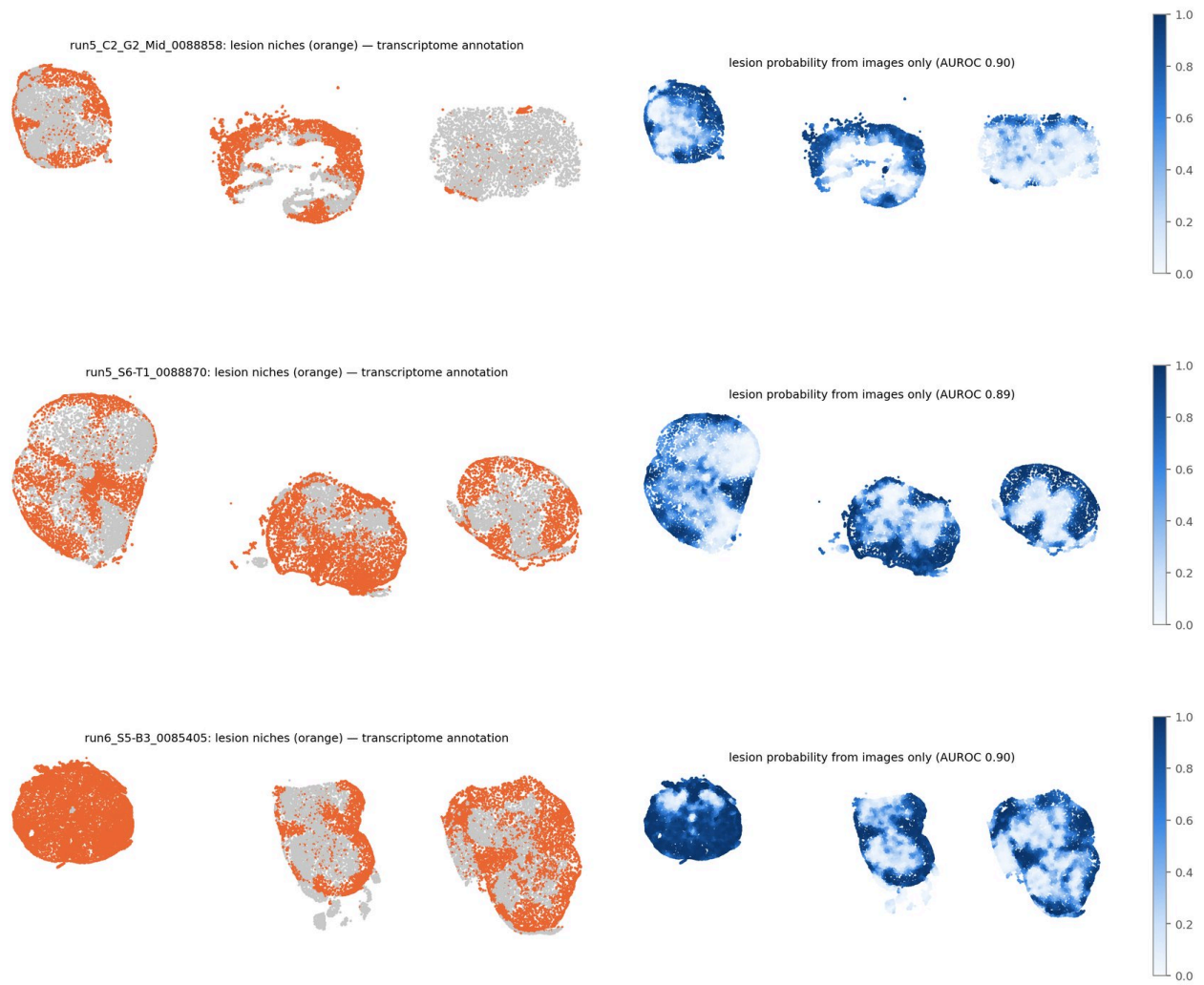
Anatomy and lesion state — images add to the transcriptome for anatomy (0.68 → 0.73):

	task	inputs	balanced acc.	chance (balanced)
0	anatomical region	images (cell + neighbourhood)	0.625	0.100
1	anatomical region	transcriptome cell-type composition	0.683	0.100
2	anatomical region	both	0.730	0.100
3	lesion state	images (cell + neighbourhood)	0.535	0.143
4	lesion state	transcriptome cell-type composition	0.648	0.143
5	lesion state	both	0.665	0.143



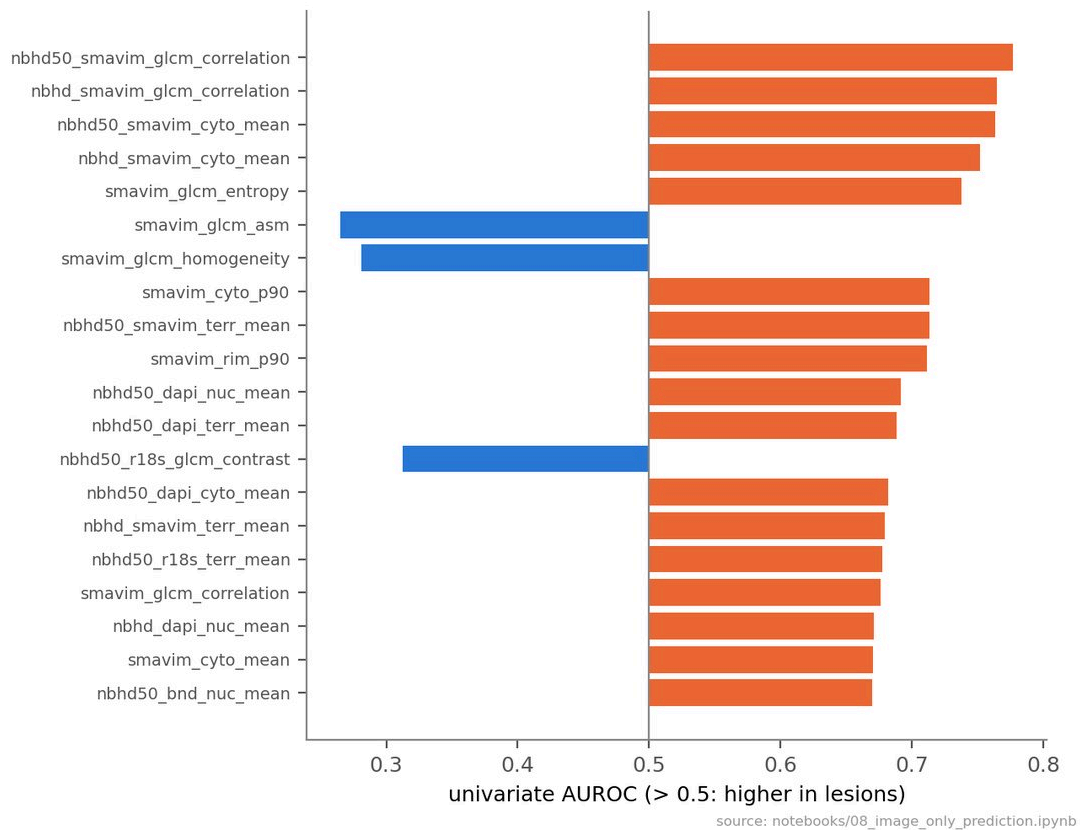
results/08_image_only_prediction/anatomy_lesion_prediction.png

Lesion maps from images alone (held-out animals; AUROC 0.90, transcriptome composition 0.92, both 0.93):



source: notebooks/08_image_only_prediction.ipynb

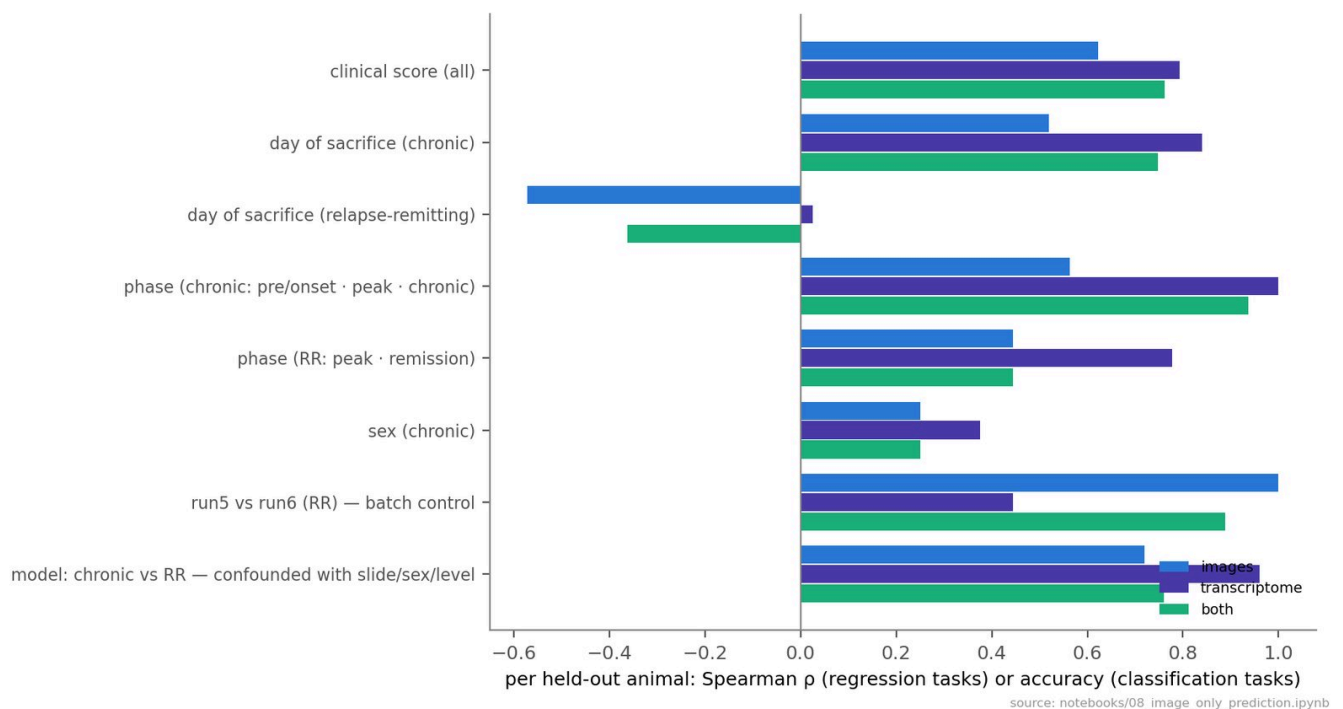
results/08_image_only_prediction/lesion_maps_images_vs_annotation.png



results/08_image_only_prediction/lesion_top_image_features.png

Animal metadata — images track score and chronic timepoint but stay below the transcriptome; they identify the **imaging run** perfectly (strong batch signature — important for adding runs 1–3):

task	metric	input	images	transcriptome	both
clinical score (all)	Spearman ρ (per animal)		0.62	0.79	0.76
day of sacrifice (chronic)	Spearman ρ (per animal)		0.52	0.84	0.75
day of sacrifice (relapse-remitting)	Spearman ρ (per animal)		-0.57	0.03	-0.36
phase (chronic: pre/onset · peak · chronic)	accuracy (per animal)		0.56	1.00	0.94
phase (RR: peak · remission)	accuracy (per animal)		0.44	0.78	0.44
sex (chronic)	accuracy (per animal)		0.25	0.38	0.25
run5 vs run6 (RR) — batch control	accuracy (per animal)		1.00	0.44	0.89
model: chronic vs RR — confounded with slide/sex/level	accuracy (per animal)		0.72	0.96	0.76



results/08_image_only_prediction/animal_metadata_prediction.png

Caveats

- CD45 component undetectable → no immune-membrane readout from ch1.

- α SMA and vimentin share a channel; 18S drew ~95 % of masks (18S distribution features partly by construction).
- Lesion niches are transcriptome-defined, favouring the transcriptome in lesion comparisons.
- 25 animals; only 2 CFA control pieces in runs 5/6 (runs 1–3 were added later, notebooks 10–19).

Next steps

1. ~~Validate the 18S texture severity marker on runs 1–3~~ done (notebook 13): partly replicated.
2. Disentangle vimentin vs α SMA: post-Xenium immunofluorescence with separate antibodies (see the verdict document).
3. ~~Perivascular T-cell 18S polarity~~ withdrawn (notebook 19).
4. Use the neuropil-loss index and 18S per cell as covariates in the RRMAP2 lesion analyses.
5. Runs 1–3: image features carry a strong run signature — compare within run / batch-correct before pooling.